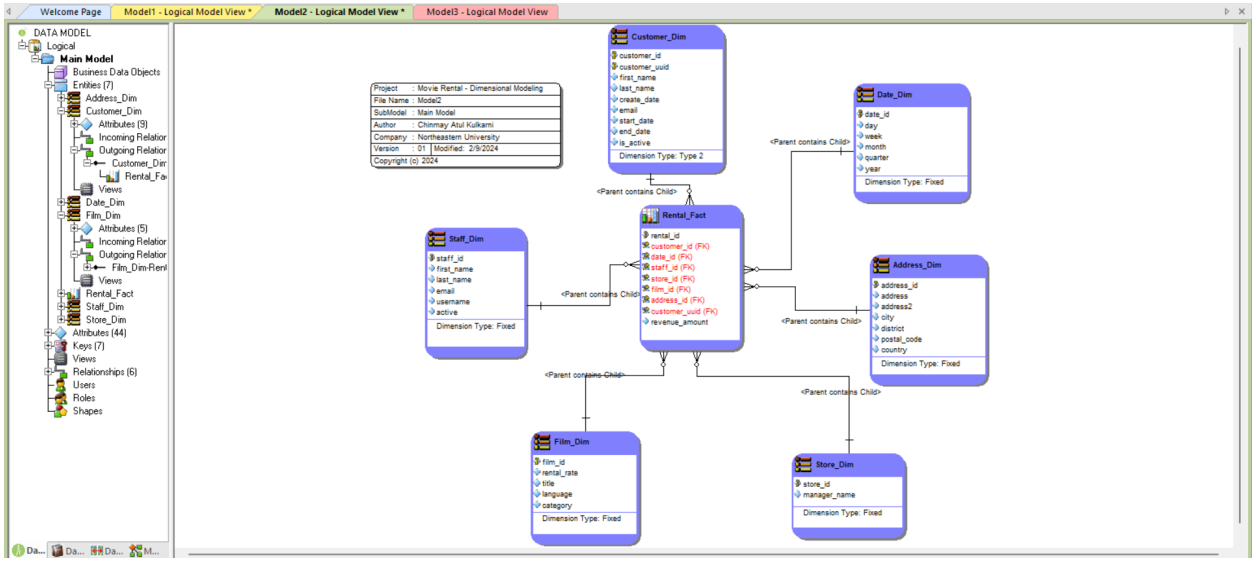
Assignment - Sakila Dimensional Modeling →

Saikla Dimensional Modeling ER-Studio Screenshot:



All these tables are connected to the Rental Fact table through their primary keys acting as foreign keys in the Rental Fact table. This star schema design allows for complex analytical queries to be performed across various dimensions such as time, location, customer, staff, and product (film)

Facts →

- <u>Rental_Fact</u>
- rental_id
- customer_id (FK)
- date _id (FK)
- staff_id(FK)
- store_id(FK)
- film_id(FK)
- address_id(FK)
- customer_uuid(FK)
- revenue_amount

Dimensions →

1) <u>Customer_Dim</u> **(SCD Type 2 Dimension)**
   Attributes:-
   - customer_id
   - Customer_uuid (durability_key)
   - first_name
   - last_name
   - create_date
   - email
   - start_date

- end_date
- is_active

2) <u>Staff_Dim</u>
   Attributes:-
   - staff_id
   - first_name
   - last_name
   - email
   - username
   - active

3) <u>Store_Dim</u>
   Attributes:-
   - store_id
   - manager_name

4) <u>Date_Dim</u>
   Attributes:-
   - date_id
   - day
   - week
   - month
   - quarter
   - year

5) <u>Address_Dim</u>
   Attributes:-
   - address_id
   - address
   - address_2
   - city
   - district
   - postal_code
   - country

6) <u>Film_Dim</u>
   Attributes:-
   - film_id
   - rental_rate
   - title
   - language
   - category

Questions →

Granularity -
Granularity describes the level of depth or refinement in a system, dataset, or analysis. It refers to how finely something is divided or measured. According to our business requirements, the model's granularity is day/week/month/quarter/year. The least being is a **day**.

Q1. How many customers were onboarded this month/Week/QTR/YEAR?
Ans - It is necessary to use the Customer_Dim and Date_Dim fields to respond to the business requirement of finding the total number of customers onboarded in a given period (month, week, quarter, or year).

Q2. More frequently rented movies?
Ans - To find the most frequently rented movie, analyze the Rental_Fact table to count rentals per movie, then use the Film_Dim table to identify the movie. This involves aggregating rental transactions by film_id, determining which movie has the highest count

Q3. How much revenue did each store generate?
Ans - We will total up the rental transaction revenue from each store_id in the Rental_Fact database to determine how much money each store made. In this step, the revenue_amount for each store is added together, or if revenue_amount is derived, it is calculated using rental_rate from the Film table.

Q4. Compare revenue between each store?
Ans - You will use the Rental_Fact table, which has the revenue and the store_id for each transaction, along with the Store_Dim table to identify the specific stores, to compare the revenue between each store using your dimensional model.

Q5. DVD Rentals this month vs last?
Ans - Generally, you will utilize the Date_Dim table along with the Rental_Fact table to solve the business question of comparing DVD rentals this month versus last month.

Q6. Most frequent renting customers?
Ans - As the Rental_Fact table keeps track of all rental transactions, you could look at it to find the clients who rent most frequently. You can identify which customers are renting the most by using the customer_id in the Rental_Fact database.